AIAA Applied Aerodynamics
13th Hover Invited Sessions: SciTech 2026

January 12-16, 2026, Orlando, Florida

The AIAA Applied Aerodynamics Rotorcraft Simulation Discussion Group will host multiple special sessions at SciTech 2026. These are a continuation of the previous SciTech special hover sessions that were held from 2014 to 2025. Organizers seek high quality papers addressing the challenges of hover simulations including accurate and rapid rotor performance prediction, aeroelastic windup, boundary layer transition, flow separation prediction, wake capture, and aerodynamic interactional effects.

Papers that validate methodologies are encouraged to use isolated rotor test data acquired for the Hover Validation and Acoustic Baseline (HVAB) rotor. Data for validation can be found at https://rotorcraft.arc.nasa.gov/HVAB and include performance, thermography, shadowgraphy, photogrammetry, PIV, and pressure and airloads.

Organizers also seek hover and low speed flight calculations for multi-rotor eVTOL or UAV rotorcraft. Of particular interest are shareable rotor and airframe geometry with test data for future common simulations with the HPW community.

Elements of the Hover Focus Problem (HFP) are desired to demonstrate predictive capability of hover in/out of ground effect, groundwash, head and tail wind impacts, and helicopter trim while descending to the ground. Details of the HFP can be found at https://www.aiaa-hpw.org/real-hover-problem.

Call for AIAA SciTech 2026 papers is available at: https://www.aiaa.org/SciTech/call-for-content. Participants will need to upload abstracts by May 22, 2025. Abstracts should clearly detail the focus area of your hover investigations. When submitting through AIAA's conference website for SciTech-2026, please choose: Topic -> "Applied Aerodynamics", Sub-Topic ->"Special Session: Rotor-in-Hover Simulation Sessions". This will ensure that your submission is tagged correctly.

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